ParaThink: Shared Context, Parallel Thinking and Expression

Shared state inputs / Parallel thinking and expression / One playback owner

Incoming user audio

Nemotron Streaming ASR
Online transcript + prefix version

Transcript updates

Currently Available Dialogue Context + Runtime State
Transcript + generated history / Layer 2 CoT / SoulX-Duplug user states
Backchannel timing suggestions / Layer 1 output / Listen–Speak state
Interruption events / generated, submitted, canceled and played records

Read / States

Audio

SoulX-Duplug User-State Model
Audio + transcript
Five native user states
Future: CoT conditioning

Continuous audio observation
States reach Layer 1 via Context

Read / Timing

Layer 0 Backchannel Timing
System backchannel
Timing prediction

System backchannel

Read / Output

Layer 1 Expression
Read all five user states
Reply after user_complete
Request thinking

Request

Speech text

Read / CoT

Layer 2 Async Thinking
Publish CoT only
No turn-action advice
Update one active task

CoT shared with SoulX-Duplug + Layer 1
Read at legal input boundaries

Text FIFO
Submitted speech text / system backchannel, in order
Chunks carry output_id + chunk_id; accept active output only

Consume in order

Valid user_nonidle
During active output: stop
Bypass FIFO / no LLM wait

Output Runtime: Streaming TTS + Player
Single arbitration point / single owner of audible output
Priority stop: halt audio + cancel pending output
Reject late text / audio carrying an invalid output_id

Audio heard by the user: never revoked

Actual Listen / Speak state + interruption events

On a valid stop: execution in the same output runtime

1. Close old output_id
Block stale submissions

2. Halt + cancel
Cancel queued / buffered output
Discard late text / audio

3. Closure acknowledged
Publish Listen + interruption
Keep generated history

4. New Layer 1 reply
Read matching user_complete
Fresh output_id; new text

Ordinary replies begin after Layer 1 reads user_complete in the current turn; no separate reply call.

Target architecture; integration pending. CoT conditioning requires training; its trigger and output encoding remain open.